

NSDF-Data student guide 1: setting up your computer

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This guide gets one student from nothing to a working analysis environment on **Windows, macOS or Linux**. It takes about an hour, mostly downloads. Do it once. The next guide, *first analysis*, uses what is set up here.

You will need: a GitHub account, a terminal, and about 2 GB of free disk space.

1. Install the tools

Use the same tools on every system.

Tool	Why	Get it
Git	download and track the code	git-scm.com (Windows: take the defaults)
Miniforge (conda)	Python with the right packages	github.com/conda-forge/miniforge
VS Code (optional)	editor, notebook viewer	code.visualstudio.com

Windows. Install Git for Windows, then Miniforge. Open *Miniforge Prompt* from the Start menu for every command in this guide. **macOS.** Install Miniforge (`brew install --cask miniforge` if you use Homebrew), then open *Terminal*. Install Git with `xcode-select --install` if it is missing. **Linux.** Install Git and Miniforge from your package manager or the links above. Use any terminal.

Check the tools work:

```
git --version
conda --version
```

2. Get the code

Pick a folder for the project, for example `Documents/Research` on Windows or `~/Research` on macOS and Linux. Then:

```
git clone https://github.com/villano-lab/NSDF-Data.git
cd NSDF-Data
git checkout -b student-<yourname>
```

The last line makes a personal branch. **Never work on master.** You will commit your own notebooks to your branch, not to `master`.

3. Create the Python environment

The lab uses one environment, called `darkmatter_cli_env`, with Python 3.10 and pinned package versions. Create it with:

```
conda create -n darkmatter_cli_env python=3.10 -y
conda activate darkmatter_cli_env
python -m pip install nsdf-dark-matter==0.3.0 nsdf-dark-matter-cli==0.3.1 numpy==2.2.6 matplotlib
```

Use the same `conda activate` line each time you open a new terminal.

Do not create the environment with `conda env create -f environment.yml`. That file was exported on a Mac and pins macOS-only builds, so it fails on Windows and Linux. The package versions above match it.

4. Check that it works

Run the unit tests. They use synthetic data, so they need no download:

```
cd python
python -m pytest -q
cd ..
```

You should see all tests pass (the last line reads `52 passed`, or more if the library has grown). If a test fails, stop and ask before changing anything.

Check the data tool:

```
nsdf-cli version
```

It should print `NSDF Dark Matter CLI: 0.5.0` or later.

5. Set up Jupyter

Notebooks in `R76/analysis_notes/` use the kernel `darkmatter_cli_env`. Register it once:

```
python -m ipykernel install --user --name darkmatter_cli_env
```

Start Jupyter from the repository folder with `jupyter lab` (or open the `.ipynb` in VS Code and choose the kernel `darkmatter_cli_env`).

6. Keep your work safe

- **Save your work on your branch:** `git add <file>`, `git commit -m "..."`, `git push -u origin student-<yourname>`.
- **Do not commit** files under `~/idx/` (raw data, not in the repository) or large outputs.
- **Running `07221203_2025_dump1_noise.ipynb` rewrites `archives/good_noise.h5`.** Before you run it, ask whether you should, or run a copy of the notebook.
- **Update before you start a session:** on your branch, run `git fetch` and `git merge origin/master`.

Done when

- `python -m pytest -q` in `python/` passes.

- `nsdf-cli` version prints a version.
- `jupyter lab` opens and the notebook kernel is `darkmatter_cli_env`.

Next: *Student guide 2: your first analysis*.

Session Info

Guide version: 1, 5 October 2026. Questions go to the project lead.

Key links

- Notes site: <https://villano-lab.github.io/NSDF-Data/>
- Repository: <https://github.com/villano-lab/NSDF-Data>
- Conda-forge Miniforge: <https://github.com/conda-forge/miniforge>